第9次作业

习题4：4-1 解：(1) 周期为6.

则 $a_k=\frac{1}{N}\sum_{n=\langle N\rangle} x[n]\,e^{-jk(\frac{2\pi}{N})n}$

$=\frac{1}{6}\left(e^{0}+e^{-jk\frac{\pi}{3}}+e^{-jk\frac{2\pi}{3}}+e^{-jk\pi}\right)$

$=\frac{1}{6}\left(1-2j\sin\frac{k}{3}\pi+(-1)^k\right)$

则 $|a_k|=\frac{1}{3}\left|\cos\frac{k\pi}{2}+\cos\frac{k\pi}{6}\right|$，相角为 $-\frac{k\pi}{2}$.

(3) $x[n]=\sum_{m=-\infty}^{\infty}\{2\delta[n-4m]+4\delta[n-1-4m]\}$ 周期为4.

$a_k=\frac{1}{4}\sum_{n=0}^{3}\left(2e^{0}+4e^{-jk\frac{\pi}{2}}\right)=\frac{1}{2}+\cos\frac{k\pi}{2}-j\sin\frac{k\pi}{2}$.

则 $|a_k|=\sqrt{\left(\frac{1}{2}+\cos\frac{k}{2}\pi\right)^2+\sin^2\frac{k}{2}\pi}=\sqrt{\frac{5}{4}+\cos\frac{k\pi}{2}}$

相角为：$\arctan\dfrac{-\sin\frac{\pi}{2}k}{\frac{1}{2}+\cos\frac{k}{2}\pi}=\begin{cases}0, & k=4m\\ -\arctan 2, & k=4m+1\\ \pi, & k=4m+2\\ \arctan 2, & k=4m+3\end{cases}$

(5) $a_k=\frac{1}{4}\left(e^{0}+\left(1-\frac{\sqrt{2}}{2}\right)e^{-jk\frac{\pi}{2}}+0+\left(1-\frac{\sqrt{2}}{2}\right)e^{-jk\frac{3\pi}{2}}\right)$

$=\frac{1}{4}\left(1+\left(1-\frac{\sqrt{2}}{2}\right)\cdot\left(\cos\frac{\pi}{2}k+\cos\frac{3\pi}{2}k-j\sin\frac{\pi}{2}k-j\sin\frac{3\pi}{2}k\right)\right)$

k为奇时：$a_k=\frac{1}{4}$

k为偶时：$a_k=\frac{1}{4}\left[1+(2-\sqrt{2})\cos\frac{\pi}{2}k\right]$

则 $a_k=\frac{1}{4}\left[1+(2-\sqrt{2})\cos\frac{\pi}{2}k\right]$

$|a_k|=\left|\frac{1}{4}\left[1+(2-\sqrt{2})\cos\frac{k\pi}{2}\right]\right|$，相角为$0^\circ$

4-2 解：(1) $x[n]=\sum_{k=\langle N\rangle} a_k e^{jk(\frac{2\pi}{N})n}$

$=e^{0}+\sqrt{2}e^{j\frac{\pi}{4}n}-e^{j\frac{\pi}{2}n}+0-e^{j\pi n}-\sqrt{2}e^{j\frac{5\pi}{4}n}+e^{j\frac{3\pi}{2}n}+0$

$=1-(-1)^n+\sqrt{2}\left(\cos\frac{\pi}{4}n+j\sin\frac{\pi}{4}n-\cos\frac{5\pi}{4}n-j\sin\frac{5\pi}{4}n\right)$
$-\cos\frac{\pi}{2}n-j\sin\frac{\pi}{2}n+\cos\frac{3\pi}{2}n+j\sin\frac{3\pi}{2}n$.

$=\begin{cases}2+2\sqrt{2}\left(\cos\frac{\pi}{4}n+j\sin\frac{\pi}{4}n\right)-2j\sin\frac{\pi}{2}n, & n为奇\\ 0, & n为偶.\end{cases}$

则 $0\le n\le 7$ 时 $x[1]=-4$，$x[3]=4j$，$x[5]=-4j$，$x[7]=4$，其余为0

(3) 周期为4.

$x[n]=1\times e^{0}+1\times e^{j\frac{\pi}{2}n}+0+1\times e^{j\frac{3\pi}{2}n}=1+2\cos\frac{\pi}{2}n.$

(5) $x[2n]=\sum_{k=0}^{8}a_k e^{jk\frac{\pi}{2}}$ 4

由于 $a_k=-a_{k-4}$，$e^{jk\frac{\pi}{2}}=e^{j(k+4)\frac{\pi}{2}}$

故 $x[2n]=0$.

则 $x[n]=\begin{cases}0, & n为偶数\\(-1)^{n-1}, & n为奇数\end{cases}$

4-3. 解：(1) 由 $a_k=\frac{1}{N}\sum_{n=<N>}x[n]e^{-jk(\frac{2\pi}{N})n}$

则 $a_{2k}=\frac{1}{N}\sum_{n=<N>}x[n]e^{-jk\frac{4\pi}{N}n}$.

由于 $x[n]=-x[n+\frac{N}{2}]$，$e^{-jk\frac{4\pi}{N}n}=e^{-jk\frac{4\pi}{N}(n+\frac{N}{2})}$

故 $a_{2k}=0$，证毕.

(2) $a_{4k}=\frac{1}{N}\sum_{n=<N>}x[n]e^{-jk\frac{8\pi}{N}n}$

由于 $x[n]=-x[n+\frac{N}{4}]$，$e^{-jk\frac{8\pi}{N}n}=e^{-jk\frac{8\pi}{N}(n+\frac{N}{4})}$

则 $a_{4k}=0$. 证毕.

4-4 解：(1) 证明：$a_k=\frac{1}{N}\sum_{n=<N>}x[n]e^{-jk\frac{2\pi}{N}n}$

$=\frac{1}{N}\sum_{n=<N>}x[n]\cos k\frac{2\pi}{N}n-j\frac{1}{N}\sum_{n=<N>}x[n]\sin k\frac{2\pi}{N}n$

则 $b_k=\frac{1}{N}\sum_{n=<N>}x[n]\cos k\frac{2\pi}{N}n$，$c_k=-\frac{1}{N}\sum_{n=<N>}x[n]\sin k\frac{2\pi}{N}n$.

易见：$b_{-k}=b_k$，$c_k=-c_k$

则 $a_{-k}=b_{-k}+c_{-k}=b_k-jc_k=a_k^*$. 证毕

(2) $c_{\frac{N}{2}}=-\frac{1}{N}\sum_{n=<N>}x[n]\sin\pi n=0$.

则 $a_{\frac{N}{2}}=b_{\frac{N}{2}}+j\cdot 0$，显然为实数. 证毕

4.15. 解：(1) 证明：$x[n]y[n]=\sum_{l=<N>}a_l e^{jl\frac{2\pi}{N}n}\cdot\sum_{m=<N>}b_m e^{jm\frac{2\pi}{N}n}$

(记 $k=m+l$) $=\sum_{l=<N>}\sum_{k=<N>}a_l b_{k-l}e^{jk\frac{2\pi}{N}n}$

$=\sum_{k=<N>}\left(\sum_{l=<N>}a_l b_{k-l}\right)e^{jk\frac{2\pi}{N}n}$.

即：$x[n]y[n]\xrightarrow{FS}\sum_{l=<N>}a_l b_{k-l}$. 证毕.

(2) ① 由 $\cos\left(\frac{6\pi n}{N}\right)=\frac{1}{2}\left(e^{j3\frac{2\pi}{N}n}+e^{-j3\frac{2\pi}{N}n}\right)$

得：$b_3=\frac{1}{2}=b_{-3}$

故 $c_k=\sum_{l=<N>}a_l b_{k-l}=\frac{1}{2}(a_{k-3}+a_{k+3})$

② 由于 $\sum_{r=-\infty}^{\infty}\delta[n-rN]\xrightarrow{FS}b_k=\frac{1}{N}\sum_{r=-\infty}^{\infty}e^{-jk\frac{2\pi}{N}rN}=\frac{1}{N}\sum e^{-j2\pi kr}$

$=\frac{1}{N}$.

故 $c_k=\sum_{l=<N>}a_l\cdot\frac{1}{N}=\frac{1}{N}\sum_{l=<N>}a_l$.

(3) 由 $x[n]=\cos\frac{\pi n}{3}=\frac{1}{2}\left(e^{-j2\cdot\frac{\pi}{6}n}+e^{j2\frac{\pi}{6}n}\right)$

得：$a_2=a_{-2}=\frac{1}{2}$.

又 由定义：$b_k=\frac{1}{N}\sum_{n=<N>}y[n]e^{-jk\frac{2\pi}{N}n}$

$=\frac{1}{12}\left(e^{jk\frac{\pi}{2}}+e^{jk\frac{\pi}{3}}+e^{jk\frac{\pi}{6}}+e^0+e^{-jk\frac{\pi}{2}}+e^{-jk\frac{\pi}{3}}+e^{-jk\frac{\pi}{6}}\right)$

$=\frac{1}{12}\left(1+2\cos\frac{\pi}{2}k+2\cos\frac{\pi}{3}k+2\cos\frac{\pi}{6}k\right)$

则 $c_k=\sum_{l=<N>}a_l b_{k-l}=a_2 b_{k-2}+a_{-2}b_{k+2}$

$=\frac{1}{12}\left(1-2\cos\left(\frac{\pi}{2}k\right)-\cos\left(\frac{\pi}{3}k\right)+\cos\left(\frac{\pi}{6}k\right)\right)$

(4) 证明：由(1)可知：$\sum_{l=<N>}a_l b_{k-l}=\frac{1}{N}\sum_{n=<N>}x[n]y[n]e^{-jk\frac{2\pi}{N}n}$.

令 $k=0$，即得：$\sum_{l=<N>}a_l b_{-l}=\frac{1}{N}\sum_{n=<N>}x[n]y[n]$，证毕